Early Detection of Multimodal Fake News via Reinforced Propagation Path Generation

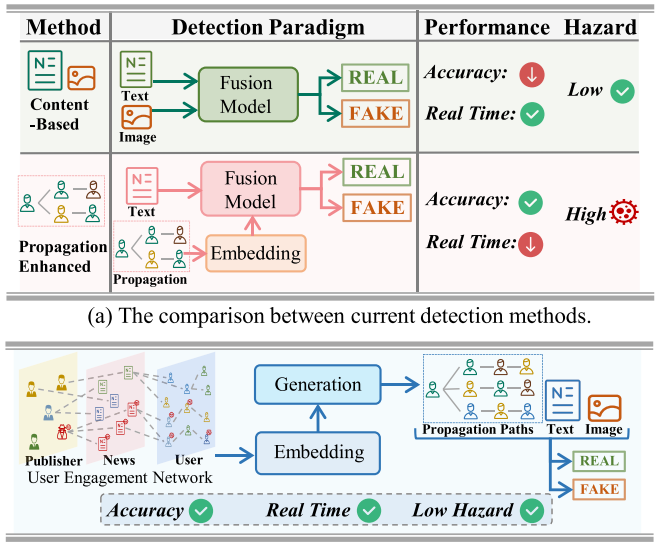
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***Abstract*—Amidst the rapid propagation of multimodal fake news across social media platforms, the detection of fake news has emerged as a prime research pursuit. To detect heightened level of meticulous fabrications, propagation paths are introduced to provide nuanced social context that enhances the basic semantic analysis of the news content. However, existing propagation-enhanced models encounter a dilemma between detection efficacy and social hazard. In this paper, we explore the innovative problem of early fake news detection through the generation of propagation paths, capable of benefiting from the extensive social context within propagation paths while mitigating potential social hazards. To address these challenges, we propose a novel Reinforced Propagation Path Generation Fake News Detection model, *RPPG-Fake*. Departing from conventional discriminative approaches, *RPPG-Fake* captures the propagation topology pattern from a heterogeneous social graph and generates the propagation paths to detect fake news effectively under a reinforcement learning paradigm. Our proposal is extensively evaluated over three popular datasets, and experimental results demonstrate the superiority of our proposal.**

***Index Terms*—Contrastive learning, fake news detection, heterogeneous graph, reinforcement learning.**

I. Introduction

THE proliferation of fake news has become a pressing concern in contemporary society, driven by the exponential growth of online social media platforms [1]. This phenomenon not only undermines the veracity of information but also poses significant social hazards due to its widespread dissemination. Conventional methods for detecting fake news [2], [3], [4] typically focus on capturing the semantics from multimedia news, including images or text, via deep modeling architectures, as depicted in Fig. 1(a). In the pursuit of effective evasion of detection [5], recent fake news is often meticulously fabricated by skilled malicious publishers, rendering them challenging to detect methods that rely solely on semantic patterns [6].



(a) The comparison between current detection methods.

(b) The proposed propagation simulation framework.

Fig. 1. Comparison of different detection methods.

To enhance the detection capabilities against meticulous fabrication, the propagation path is introduced as a complement to conventional semantic analysis methods [7], [8]. The propagation path encompasses the intricate dynamics of user interactions during the dissemination of news across social platforms [9], reflecting both temporal and structural features inherent to the news spreading process. The propagation paths associated with fake news often exhibit distinct characteristics compared to real news items. Specifically, these paths tend to manifest as wider and deeper networks, involving a higher prevalence of social bots and automated accounts [10], [11]. Recognizing these distinct characteristics can serve as vital clue for fake news detection. The integration of propagation paths in fake news detection augments the available social context, thereby facilitating more effective detection.

The integration of propagation paths in the detection of well-fabricated news holds promise for enhancing detection accuracy. However, these approaches grapple with inherent dilemma concerning its **detection efficacy** and the potential **social hazards** associated with extensive propagation paths.

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Our code is publicly available in https://github.com/LitianD/RPPG-Fake/.

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The abundant social context within extensive propagation paths aids in of detecting fake news effectively. Conversely, the presence of adequate propagation paths also signifies that the fake news might have already permeated in social platforms deeply, thereby exacerbating social hazards, as displayed in Fig. 1(a). Efforts aimed at detecting widespread dissemination fake news, which has caused societal damage, face inherent constraints indicative of ineffectual endeavor. To tackle this dilemma and identify fake news at an early stage, various social context information has been leveraged. These include user profiling [12], [13], assessment of publisher credibility [14], and analysis of user comments [15]. Nonetheless, the incorporation of extra auxiliary information demands substantial time and labor investments, limiting its applicability in data-sparse scenarios. Moreover, the propagation topology structures and temporal features within the limited propagation paths may not suffice for comprehensive detection. Consequently, the reliance solely on propagation-enhanced detection methods may leave significant blind spots, allowing fake news to evade detection until after substantial societal hazards have been inflicted from widespread dissemination.

The integration of propagation paths improves the detection performance, albeit accompanied by associated social hazards stemming from the proliferation of fake news. This leads us to ponder: *Can we leverage the abundant social context in the propagation paths to enhance the early-stage detection before widespread dissemination?* In contrast to current discriminative models, our motivation lies in forecasting and generating the forthcoming propagation paths based on the multimodal content before the proliferation of fake news. As depicted in Fig. 1(b), our aim is to capture the propagation topology pattern within the user engagement network, thereby facilitating the generation of propagation paths, thus enhancing the capacity for early detection of emergent news. This generation-based strategy does not necessitate additional social context during news spreading and is anticipated to generate rich propagation features to enhance detection capabilities while migrating the social hazards from news spreading.

However, generating propagation paths to simulate propagation process encounters significant challenges, primarily due to three key reasons. First, the effective and reasonable user encoding is the pre-requisite for path generation task. The propagation process in the whole social platform contains large unlabeled interactions among various roles involved in the news dissemination process, making it non-trivial to model the whole propagation process and learn the user representation effectively. Second, the sheer size of the user engagements makes it intractable to generate the propagation paths in discrete space that adequately captures the pattern of real news spreading. Finally, modeling structural and temporal features from generated propagation paths simultaneously while enhancing the detection performance of multimodal fake news poses significant challenges.

In this paper, we propose a novel *Reinforced Propagation Path Generation Fake News Detection* (*RPPG-Fake*) model for multimodal fake news detection. Diverging from the prevailing discriminative detection methods, we propose to simulate realistic propagation paths based on news content at the early detection stage. To generate reasonable user embedding, the global user interaction process is modeled as the heterogeneous graph. In particular, the self-supervised dual-contrastive learning mechanism is designed to capture the propagation topology feature and heterogeneous relation features in heterogeneous social graph. The generation process of propagation path is facilitated by a novel reinforcement learning module. This module aids in simulating user engagements temporally, which contributes to conducting an efficient exploration from steer size user searching space. Moreover, a temporal-aware detection module is proposed, wherein a dynamic graph neural network is designed to encode temporal information from generated propagation paths, facilitating enhanced performance for final detection. The *RPPG-Fake* undergoes comprehensive evaluation across three datasets, with the experimental results illustrating the efficacy and superiority of our proposal.

Our contributions are summarized as follows:

- To the best of our knowledge, we are the first to explore the challenge of early fake news detection by generating propagation paths, which leverage the advantages of extensive social context while mitigating social hazards.
- We propose a novel *RPPG-Fake* model to generate reasonable propagation paths via reinforcement learning, and further integrate propagation topology structure and temporal depth information through a temporal-aware fake news detection module.
- Comprehensive experiments across three real-world datasets demonstrate that our proposal consistently surpasses state-of-the-art baselines in fake news detection.

II. Problem Formulation

Definition 1 (Fake News Detection): In the context of fake news detection, a news item is represented as $n = \{n^T, n^V\}$, where n^T refers to the textual content and n^V refers to the visual content. The collection of all news items is denoted as $N = \{n_1, n_2, \ldots, n_{|N|}\}$. We also define the set of users as $U = \{u_1, u_2, \ldots, u_{|U|}\}$, and the set of publishers as $P = \{p_1, p_2, \ldots, p_{|P|}\}$. For a specific news item, the propagation paths at a given time t are depicted as $G_t = (U_t, E_t)$, where each edge $e_i \in E_t$ represents a connection between two interacting users. The complete set of propagation paths during all spreading times is denoted as $\mathcal{G} = \{G_1, G_2, \ldots, G_T\}$, where T denotes the final spreading time. The primary goal of fake news detection is to model a probability distribution $\mathbb{P}(y|n, \mathcal{G})$, where $y = 0$ indicates fake news and $y = 1$ stands for real.

Definition 2 (Fake News Early Detection): During the early detection stage, the available propagation paths are limited and may even be nonexistent. In this condition, the absence of user interactions for a specific news item n is represented by the sequence $\mathcal{G} = \{\emptyset\}$. Our objective is to generate propagation paths and effectively distinguish between the real news and fabricated news with insufficient propagation paths at an early detection stage.

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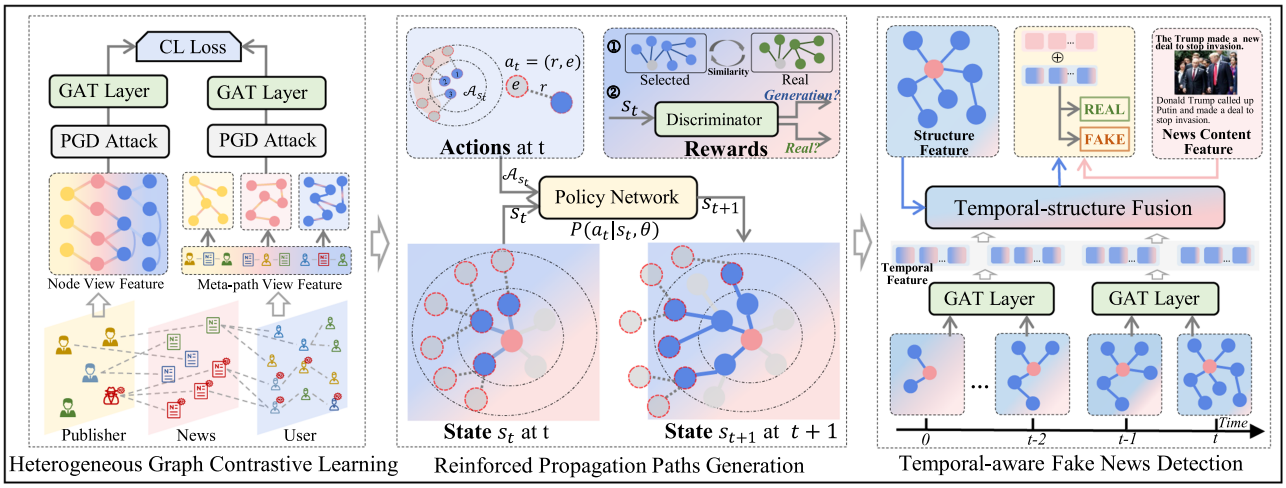


Fig. 2. The overview of proposed *RPPG-Fake* model. The model includes: (1) Heterogeneous Graph Contrastive Learning for feature extraction, (2) Reinforced Propagation Paths Generation for selecting optimal propagation actions, and (3) Temporal-aware Fake News Detection using fused structural, content, and temporal features. Yellow nodes are publishers, red nodes are news items, and blue nodes are users.

III. Methodology

The proposed *RPPG-Fake* model framework is depicted in Fig. 2, which aims to predict the label y by leveraging multi-modal news content and generated propagation paths. Mathematically, our proposal is formulated as:

$$\mathbb{P}(y \mid n, \mathcal{G}) = \underbrace{\mathbb{P}(y \mid n, \mathcal{G}')}_{\text{Temporal Aware Detection}} \underbrace{\mathbb{P}(\mathcal{G}' \mid \mathcal{G}, \mathcal{H})}_{\text{Reinforced Generation}} \underbrace{\mathbb{P}(\mathcal{H} \mid P, N, U)}_{\text{Contrastive Learning}} \tag{1}$$

where $\mathcal{H}$ denotes as the constructed heterogeneous graph. $\mathcal{G}'$ represents the generated propagation paths. The *RPPG-Fake* model consists of three modules as below:

Heterogeneous Graph Contrastive Learning: This module focuses on constructing a social heterogeneous graph $\mathcal{H}$ to model the global user interaction process. Specially, a dual-view contrastive learning mechanism is designed, the goal is to capture the propagation topology patterns effectively.

Reinforced Propagation Paths Generation: This module simulates the news spreading process and generates propagation paths through a reinforcement learning framework. The generation process of propagation paths $\mathcal{G}'$ is based on the user embedding in $\mathcal{H}$ and early stage propagation $\mathcal{G}$, which could be applied even if no propagation paths exist.

Temporal-aware Fake News Detection: In order to consider both the propagation structure and temporal information, the proposed temporal-aware detection module aims to model dynamic propagation graphs. By incorporating structure and temporal features into the news content, this module enhances the accuracy of fake news detection.

A. Heterogeneous Graph Contrastive Learning

Previous propagation enhanced models [16], [17] often focus on incorporating individual news propagation paths as supplementary information to identify fake news. However, these methods are limited in capturing the global propagation features across all news on social platforms. The dissemination of news on social platforms involves various roles (such as publishers, users, and news content) that are interconnected in diverse relationships, offering valuable insights for understanding the intricate propagation interactions. To comprehensively model the spreading of news on social platforms, we have constructed a heterogeneous graph denoted as $\mathcal{H} = (\mathcal{V}, \mathcal{E}, \mathcal{U})$, where $\mathcal{V}$ represents the node set, $\mathcal{E}$ is the edge set with edges belonging to $\mathcal{V} \times \mathcal{V}$, and $\mathcal{U}$ is the embedding feature set. Due to the social heterogeneous graph containing a large number of unlabeled interaction data, we introduce a dual-view self-supervised contrastive mechanism to learn the topology structure feature and heterogeneous relationship feature.

1) Node View Contrastive Learning: To capture the global topology structure feature across social platforms effectively, we propose a contrastive learning mechanism based on node view. The first step is to generate positive and negative node pairs for every node in the graph $\mathcal{H}$. Inspired by PGD algorithm [18], we introduce a perturbation δ into node features to create more challenging instances for both positive and negative pairs in contrastive learning. The perturbed user node feature is denoted as $\widetilde{\mathbf{u}} = \mathbf{u} + \delta$. Then positive and negative node sets are constructed. For each node $v_i \in \mathcal{V}$, two sets of nodes, $\mathcal{V}_i^p$ and $\mathcal{V}_i^n$, are defined to contain the top K_1 highest and lowest cosine similarities with the center node v_i [19]. Subsequently, the Graph Attention Network (GAT) encoder [20] is utilized to obtain the node embedding:

$$\begin{aligned} \alpha_{ik} &= \frac{\exp(\mathbf{a}_1^{\mathrm{T}} \dot{\mathbf{u}}_k)}{\sum_{k \in \mathcal{N}(i)} \exp(\mathbf{a}_3^{\mathrm{T}} \dot{\mathbf{u}}_k)}, \dot{\mathbf{u}}_k \\ &= \mathrm{ReLU}\left(\left[\mathbf{W}_1 \widetilde{\mathbf{u}}_k^{L-1}; \mathbf{W}_1 \widetilde{\mathbf{u}}_i^{L-1}\right]\right) \\ \widetilde{\mathbf{u}}_i^L &= \sigma\left(\sum_{u_k \in \mathcal{N}_i} \alpha_{ik} \mathbf{W}_1 \widetilde{\mathbf{u}}_k^{L-1}\right), \end{aligned} \tag{2}$$

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where $[;]$ is the concatenation operation, σ is the activation function, $\mathbf{W}_1$ is trainable weight matrix, $\mathbf{a}_1$ is the weight vector. α_{ik} is the attention coefficient, L is the layer of GAT. We further simply the notation of last layer node embedding $\widetilde{\mathbf{u}}_i^L$ as $\hat{\mathbf{u}}_i$. The positive pair $(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j)$ is formed for $v_j \in \mathcal{V}_i^p$, while the negative pair is created for $v_j \in \mathcal{V}_i^n$.

The node-view contrastive loss is calculated using the positive and negative sample sets for node v_i as follows:

$$\mathcal{L}_i^{node} = -\log \frac{\sum_{j \in \mathcal{V}_i^p} \exp\left(sim\left(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j\right)/\tau_1\right)}{\sum_{k \in \mathcal{V}_i^n} \exp\left(sim\left(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_k\right)/\tau_1\right)}, \tag{3}$$

where $sim(\hat{\mathbf{u}}_i, \hat{\mathbf{u}}_k)$ is the consine similarity between the node embedding (v_i, v_j) and τ_1 is the temperature parameter [21].

2) Meta-Path View Contrastive Learning: Contrastive learning in node view aims to integrate the neighbor structure and global propagation information, but it tends to neglect the diverse semantic relationships among various roles in news dissemination. Introducing meta-paths can effectively capture the semantic connections between nodes from a high-order perspective. By defining meta-paths such as *PNU*, *NUN*, and *UNU*, we can better characterize the inter-relatedness across social platforms.

Considering a meta-path Φ_l, the random walk with restart (RWR) guided by Φ_l is employed to select neighbors based on the meta-path and generate the corresponding semantic graph $\widetilde{\mathcal{H}}^{\Phi_l}$. Similarly to (2), we utilize GAT to acquire node embeddings $\hat{\mathbf{h}}_i^{\Phi_l}$. For a pair of node embeddings $(\hat{\mathbf{h}}_i^{\Phi_l}, \hat{\mathbf{h}}_j^{\Phi_m})$, if $i = j$ and $l \neq m$, they are considered positive pairs. Conversely, if $i \neq j$ and $l \neq m$, they are categorized as negative pairs. The contrastive learning loss from the meta-path view is defined as follows:

$$\mathcal{L}_i^{meta} = -\log \frac{\exp\left(\text{sim}\left(\hat{\mathbf{h}}_i^{\Phi_l}, \hat{\mathbf{h}}_j^{\Phi_m}\right)/\tau_2\right)}{\sum_{k=1}^{p} \mathbb{I}_{[i \neq k]} \exp\left(\text{sim}\left(\hat{\mathbf{h}}_i^{\Phi_l}, \hat{\mathbf{h}}_k^{\Phi_m}\right)/\tau_2\right)}, \tag{4}$$

where τ_2 represents the temperature parameter within the meta-path view and $\mathbb{I}_{[i \neq k]}$ functions as an indicator with $\mathbb{I} = 1$ when $i \neq k$ and $\mathbb{I} = 0$ otherwise [22]. p denotes the quantity of nodes within each batch utilized in contrastive learning. Hence, the contrastive learning objective $\mathcal{L}^{CL}$ is defined as:

$$\mathcal{L}^{CL} = \sum_{i \in \mathcal{V}} \left[\lambda_1 \mathcal{L}_i^{node} + (1 - \lambda_1)\, \mathcal{L}_i^{meta}\right], \tag{5}$$

where $\lambda_1 \in [0, 1]$ is a weighting coefficient. Through the implementation of dual-contrastive learning, the global interaction features and heterogeneous relation features are represented effectively. This approach significantly enhances the construction of user embeddings, yielding representations that are more diverse and reasonable.

B. Reinforced Propagation Paths Generation

After the user embedding is generated by dual-view contrastive learning, we target to simulate propagation process with user embedding based on news content features. Since users in propagation paths are represented in discrete spaces, defining a differentiable model as the generator directly becomes challenging. Therefore, we employ the policy gradient approach to optimize the generator and formulate the generation of propagation paths as a reinforcement learning problem.

1) The Markov Decision Process: The user generation within the reinforcement learning framework, is formulated as a classic Markov Decision Process (MDP) represented by a 4-tuple $(\mathcal{S}, \mathcal{A}, \Upsilon, \Lambda)$. Here, $\mathcal{S}$ signifies the finite state space during the exploration of user interaction, while $\mathcal{A}$ denotes the action space for selecting the next user. The function $\Upsilon : \mathcal{S} \times \mathcal{A} \to \mathcal{R}$ serves as the reward function derived from the environment, used to evaluate the quality of the generated propagation path. Additionally, $\Lambda : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$ represents the transition probability function that determines the state transition based on the current state and the selected action.

State: In the propagation simulation process, each state $s_t \in \mathcal{S}$ is denoted by a 4-tuple (n, U_t, V_t, R_t), where U_t signifies the newly selected users at step t. V_t and R_t represent the sets of all users and connected edges up to step t, respectively. The initial state is represented as $s_0 = (n, \{u_n\}, \{u_n\}, \emptyset)$, where u_n denotes the seed node of the news item itself.

Action: The action space $\mathcal{A}_t$ for state s_t is defined as the collection of outgoing connections among users in E_t, denoted as $\mathcal{A}_t = \{(r, e) | (u_h, r, u) \in \mathcal{H}, u_h \in E_t, u \notin V_t\}$.

Reward: The reward function assesses the quality of navigated state. We define two rewards to provide feedback, the propagation path temporal reward and the propagation path global reward. Detailed elaboration can be found in Section III-B3.

Transition: Given a state of current propagation path $s_t = (n, U_t, V_t, R_t)$ and an action $a_t = \{(r, e)\}$, the transition to the next state is defined as:

$$\mathbb{P}\left(s_{t+1} = (n, U_{t+1}, V_{t+1}, R_{t+1}) | s_t, a_t\right) = 1, \tag{6}$$

where the V_{t+1} in $t+1$ step is built by adding new user nodes U_{t+1} and relations R_{t+1} from the t step V_t.

2) The Generation Policy Network: The neural policy network $\pi_\theta(s_t, a_t)$ is responsible for estimating the conditional probability distribution of actions given a state [23].

We first initialize the user embedding through the node-view and meta-view embedding from social interaction heterogeneous graph as follows:

$$\mathbf{u_i} = \text{Tanh}\left(\mathbf{W}_u^\theta \left[\hat{\mathbf{u}}_i; \sum_{l \in \Phi(u_i)} \hat{\mathbf{h}}_i^{\Phi_l}\right]\right), \tag{7}$$

where $\hat{\mathbf{u}}_i$ is the node-view embedding and $\hat{\mathbf{h}}_i^{\Phi_l}$ is the meta-view embedding with meth-path l. Then the current state s_t is represented as:

$$\mathbf{s_t} = \left([\mu_\mathbf{t}; \mathbf{v_t}; \mathbf{n}]\right), \tag{8}$$

where $[;]$ denotes the concatenation operation, $\mathbf{n}$ is the news content embedding, $\mu_\mathbf{t}$ is the vector representation of the users subgraph at step t, and $\mathbf{v_t}$ denotes the vector representation of newly arrived users at step t. The computation of $\mu_\mathbf{t}$ involves two key steps. First, it includes the initial refinement of the user embedding $\mathbf{u_h}$ within its neighborhood in V_t. Second, it encompasses the aggregation of all user nodes through an

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attention network:

$$\mathbf{u}_\mathbf{h}' = \text{Tanh}\left(\mathbf{W}_0^\theta\left[\mathbf{u_h}; \sum_{u_t\in\psi(u_h)}\mathbf{u_t}\right]\right),$$
$$\alpha_1^\theta(u_h) = \frac{\exp\left(\alpha_0^\theta\left(u_h\right)\right)}{\sum_{b\in U_t}\exp\left(\alpha_\theta^0\left(u_b\right)\right)},$$
$$\alpha_0^\theta\left(u_h\right) = \mathbf{w}^\theta \text{ELU}\left(\mathbf{W_1^\theta u_h'}\right),$$
$$\mu_\mathbf{t} = \sum_{u_h\in\mathcal{V}_t}\alpha_1^\theta(u_h)\mathbf{u}_h', \tag{9}$$

where ELU refers to the exponential linear unit and $\psi(u_h)$ denotes the neighbors of u_h. The embedding of newly arrived users $\mathbf{v_t}$ is acquired through the averaging of the entities within U_t:

$$\mathbf{v_t} = \frac{\sum_{\mathbf{u}_h\in U_t}\mathbf{u}_h}{|U_t|}. \tag{10}$$

Each available action a_i is represented as $\mathbf{a_i} = (\mathbf{r_i} + \mathbf{u_i})$, indicating the combination of two vectors. Subsequently, the policy network computes the probability of selecting an action unit a_i at state s_t:

$$\pi_\theta\left(s_t, a_i\right) = \frac{\exp\left(\mathbf{W_2^\theta}\text{ReLU}\left(\mathbf{W_3^\theta}\left[\mathbf{s}_t; \mathbf{a}_i\right]\right)\right)}{\sum_{a_j\in\mathcal{A}_t}\exp\left(\mathbf{W_2^\theta}\text{ReLU}\left(\mathbf{W_3^\theta}\left[\mathbf{s}_t; \mathbf{a}_j\right]\right)\right)}. \tag{11}$$

To accelerate the generation process, we let the agent absorb K_2 outgoing relations at one action a_t from user node, instead of only absorbing one relation [24]. We add a self-loop action to connect with itself, which means no user interaction happens. The generation is repeated until the size $|V_t|$ of generated graph arrives at threshold T. The T is the predefined maximum engagement users' number for each piece of news.

3) Optimization of Propagation Path Generation: The propagation generation process contains two types of rewards: Propagation Path Temporal Reward and Global Reward.

Propagation Path Temporal Reward: In order to effectively capture the temporal features of propagation, it is essential to simulate each search action at time step t to align with the propagation time slots. Consequently, the agent should prioritize selecting users who are actively engaged at the current time step along the real propagation path. To achieve this objective, we divide the propagation temporal process evenly and construct a temporal user engagement list to record all user actions at time step t during the real propagation process. This list, denoted as $\Omega(t) = \{u_1, u_2, \ldots, u_k | u_i \text{ acts at } t\}$, serves as a repository for user engagements at specific time steps. Mathematically, the reward function is represented as:

$$\mathcal{R}'(u_i, t) = \text{cosine}\left(\mathbf{u}_i, \frac{\sum_{u_j\in\Omega(t)}\mathbf{u}_j}{|\Omega(t)|}\right),$$
$$\mathcal{R}^T = \sum_{u_i\in\mathcal{A}_t}\mathcal{R}'\left(u_i, t\right). \tag{12}$$

where $\mathcal{R}'(u_i, t)$ is calculated based on the cosine similarity between user u_i and the average user engagement vector at time step t within $\Omega(t)$. By implementing this reward, we aim to enhance the agent's ability to optimize user selection based on temporal dynamics.

Propagation Path Global Reward: The real propagation paths are inherently created by the users themselves, making them naturally logical. It is essential that the simulated propagation paths mirror the real paths in terms of global structure. To achieve this, a discriminator function denoted as $\mathcal{D} \in (0, 1)$, is employed to differentiate between samples from real paths and those generated by the policy network.

The implementation of $\mathcal{D}$ is the same as the user subgraph modeling in (9). The optimization process involves fine-tuning both the policy network and discriminator:

$$\mathcal{R}^G = -\left(\mathbb{E}_{G_t\sim P_{\text{real}}}\left[\log\mathcal{D}(G_t)\right] + \mathbb{E}_{G_t\sim\pi}\left[\log(1-\mathcal{D}(G_t))\right]\right), \tag{13}$$

where $G_t = (U_t, E_t)$ denotes the propagation path, and P_{real} represents the distribution of the observed real propagation paths. The optimization process aims to enhance the agreement between the generated paths and the real paths by iteratively adjusting the policy network and discriminator. Hence the compound reward $\mathcal{R}^C$ is defined as:

$$\mathcal{R}^C = \lambda_2\cdot\mathcal{R}^T + (1-\lambda_2)\cdot\mathcal{R}^G, \tag{14}$$

where $\lambda_2 \in [0, 1]$ is a weighting coefficient.

C. Actor-Critic Optimization.

The critic evaluates the selected user by estimating the action value $Q(s, a)$ during propagation paths generation. A similar structural framework is applied to the policy network, as represented by (11):

$$Q_\phi(s_t, a_i) = \mathbf{W_0^Q}\text{ReLU}\left(\mathbf{W_1^Q}[\mathbf{s_t}\|\mathbf{a_i}]\right). \tag{15}$$

The critic network is trained through the Temporal Difference (TD) method [25]. It calculates the target q_t according to the Bellman [26] equation:

$$q_t = \mathcal{R}^C + \mathbb{E}_{a\sim\pi_\theta}\left[\gamma\cdot\mathcal{Q}_\phi\left(s_{t+1}, a\right)\right], \tag{16}$$

where γ is a decay factor. The learning process of the critic network involves minimizing the TD error:

$$\mathcal{L}_{critic} = \left(Q_\phi(s_t, a_t) - q_t\right)^2. \tag{17}$$

Meanwhile, the actor endeavors to maximize the expected payoff of each step with respect to the given Q-function:

$$J_{actor}(\theta) = \mathbb{E}_{a\sim\pi_\theta}\left[Q_\phi\left(s_t, a\right)\right]. \tag{18}$$

The optimization of the actor network is performed by policy gradient method [27], where the gradients of $J(\theta)$ with respect to θ are approximated as follows for each sampled trajectory:

$$\nabla_\theta J_{actor}(\theta) \simeq Q_\phi\left(s_t, a_t\right)\cdot\nabla_\theta log\pi_\theta\left(s_t, a_t\right). \tag{19}$$

D. Temporal-Aware Fake News Detection

After the augmentation propagation paths are generated by reinforcement learning, we target to learn the propagation global structure and temporal depth features from the sufficient simulation propagation information. Here we introduce a temporal-aware propagation fusion module, which contains temporal

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Algorithm 1: Training for *RPPG-Fake*, $\mathbb{P}(y \mid n, \mathcal{G})$.

Input: Social Graph $\mathcal{H} = (\mathcal{V}, \mathcal{E}, \mathcal{U})$, news dataset n
Output: Optimized fake news detection model

1 *Initialize parameters in* $RPPG - Fake$;
/* Heterogeneous graph contrastive learning */
2 **for** *sampled user nodes* **do**
3 Generate perturbation graph $\widetilde{\mathcal{H}} = (\mathcal{V}, \mathcal{E}, \widetilde{\mathcal{U}})$;
/* Dual-view self-supervised contrastive Learning */
4 Calculate $\mathcal{L}_i^{node}$ in node-view contrastive learning;
5 Calculate $\mathcal{L}_i^{meta}$ in meta-view contrastive learning;
6 Update user embedding with contrastive loss: $\mathcal{L}^{CL} = \sum_{i \in \mathcal{V}} [\lambda \mathcal{L}_i^{node} + (1-\lambda)\mathcal{L}_i^{meta}]$;
7 **end**
8 **for** *each news* $n \in \{n_1, n_2, \ldots, n_N\}$ **do**
/* Reinforced propagation path generation */
9 **for** $|V_t|$ *less than* T **do**
10 Generate state embedding: $\mathbf{s_t} = ([\mu_\mathbf{t}; \mathbf{v_t}; \mathbf{n}])$;
11 Sample actions with policy network: $\pi_\theta(s_t, a_i)$;
12 Update state s_{t+1}: $(n, U_{t+1}, V_{t+1}, R_{t+1})$;
13 **end**
14 Generate propagation paths: $G = \{G_0, \ldots, G_T\}$;
/* Temporal-aware fake news detection */
15 Generate temporal propagation path embedding: $\hat{\mathbf{H}} = \{\mathbf{h}_0 + \mathbf{p}_0, \mathbf{h}_1 + \mathbf{p}_1, \ldots, \mathbf{h}_T + \mathbf{p}_T\}$;
16 Integrate temporal and structure features as $\mathbf{H}'$;
17 Predict with propagation feature and content features: $\mathbb{P}(y \mid n, G) = \text{sigmod}(\mathbf{W}_P[\mathbf{h}'; \mathbf{n}] + \mathbf{b})$;
18 **end**

graph encoder and temporal-structure fusion to capture the structure and temporal features respectively.

1) Dynamic Propagation Graph Encoder: The generated propagation paths is formulated as $G = \{G_0, G_1, \ldots, G_{T-1}, G_T\}$. T is the generation step and $G_t = (U_t, E_t)$, where U_t is the set of all engagement users before time step t and E_t is the set of connected edges. For every time step propagation path, we apply a GAT network (4) to update the node embedding:

$$\begin{aligned} \left\{\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{|U_i|}\right\} &= \text{GAT}\left(U_t, E_t\right), \\ \mathbf{h}_i &= \text{readout}\left(\left\{\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{|U_i|}\right\}\right), \end{aligned} \tag{20}$$

where $\{\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{|U_i|}\}$ is the updated node embedding, the readout is applied by mean pool and the $\mathbf{h}_i$ is the generated temporal graph level embedding. The set of temporal propagation path embedding is $\hat{\mathbf{H}} = \{\mathbf{h}_0 + \mathbf{p}_0, \mathbf{h}_1 + \mathbf{p}_1, \ldots, \mathbf{h}_T + \mathbf{p}_T\}$, where p_i is the temporal position embedding at time step i.

2) Temporal-Structure Fusion: The informative temporal and structure information in propagation paths provides significant clues to verify fake news. In this section, we propose a temporal-structure fusion module to extract the temporal and structure features and integrate them into detection.

First, we embed the global propagation paths with GAT network to generate the structure features, which are formulated as $\{\hat{\mathbf{u}}_0, \hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_{|\hat{U}_T|}\} = \text{GAT}(U_T, E_T)$. The $\hat{\mathbf{U}}_T = \{\hat{\mathbf{u}}_0, \hat{\mathbf{u}}_1, \ldots, \hat{\mathbf{u}}_{|\hat{U}_T|}\}$ is the updated global structure features.

Then we apply a transformer based fusion module to integrate temporal and structure features:

$$\begin{aligned} \widetilde{\mathbf{H}} &= \text{LN}\left(\text{SelfAtten}\left(\hat{\mathbf{H}}\right) + \hat{\mathbf{H}}\right), \\ \widetilde{\mathbf{U}}_T &= \text{LN}\left(\text{CrossAtten}\left(\hat{\mathbf{U}}_\text{T}, \widetilde{\mathbf{H}}\right) + \widetilde{\mathbf{H}}\right), \\ \mathbf{H}' &= \text{LN}\left(\text{FeedForward}\left(\hat{\mathbf{U}}_\text{T}\right) + \widetilde{\mathbf{H}}\right)\right), \end{aligned} \tag{21}$$

where SelfAtten is the self-attention operation, CrossAtten is the cross-attention, FeedForward is the feed-forward layer and LN is the layer norm operation. The $\mathbf{H}'$ is the fused propagation features. We integrate the propagation structure features into temporal features through cross-attention operation and capture propagation path information completely.

3) Optimization of Fake News Detection: We further embed the news visual and textual content feature for detection [28]. Then the content features and propagation features are concatenated and fed into a prediction layer to make the final decision:

$$\begin{aligned} \mathbf{h}' &= \text{readout}\left(\mathbf{H}'\right), \quad \mathbf{n} = \left[\mathbf{n}^T; \mathbf{n}^V\right], \\ \mathbb{P}(y \mid n, G) &= \text{sigmod}\left(\mathbf{W}_P\left[\mathbf{h}'; \mathbf{n}\right] + \mathbf{b}\right), \end{aligned}$$

where $\mathbf{h}'$ is pooled propagation path feature and $\mathbf{n}$ is the news content feature. The readout is applied by mean pool. Finally, the cross-entropy loss is calculated as:

$$\mathcal{L} = -y \cdot log(\mathbb{P}(y)) - (1-y) \cdot log(1 - \mathbb{P}(y)). \tag{22}$$

The overview algorithm for the reinforced propagation paths simulation is presented in Algorithm 1. The procedure starts with encoding the user embedding in heterogeneous social graph through dual-view contrastive learning, as shown in lines 2-6. Subsequently, the propagation path G is generated by reinforcement learning, as illustrated in lines 9-14. Finally, the generated propagation path is embedded temporally and integrated with new content features for detection, as demonstrated in lines 15-17.

IV. Experiments

A. Experimental Settings

1) Datasets Details: We conduct experiments on three real-world datasets, Pheme [29], PolitiFact and GossipCop [30], which contain 478, 5,526 and 3,506 news, respectively. Pheme is collected from Twitter, a widely used social media platform. PolitiFact and GossipCop are collected from fact-check websites which are created to verify the authenticity of published news. We conduct data cleaning on all three datasets and Table I presents the basic information.

2) Baselines: We compare our model in two categories of baselines, a total of ten models.

The content-based fake news detection models:

- *BERT* [31] is a pretrained model to extract text features.

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TABLE I
DATASET STATISTICS SUMMARY OF PHEME, POLITIFACT, AND GOSSIPCOP DATASETS

Dataset	Pheme	PolitiFact	GossipCop
#. total news	3,506	478	5,526
#. fake news	1,142	294	1,935
#. real news	2,364	184	3,591
#. words per news	13.8	1545.8	578.6

- *MVAE* [1] comprises an encoder, a decoder, and a detector to classify fake news.
- *EANN* [32] designs a multi-task learning framework to detect fake news and classify events simultaneously.
- *CAFE* [4] aggregates unimodal features and cross-modal correlations to help detection.
- *LLAMA2* [4] aggregates unimodal features and cross-modal correlations to help detection.
- *Detect-GPT* [4] aggregates unimodal features and cross-modal correlations to help detection.

The propagation-enhanced detection methods:

- *CSI* [33] employs LSTM to encode the news content, and utilizes the group behavior of users for detection.
- *Bi-GCN* [16] uses a Bidirectional Graph Convolutional Network to learn the propagation patterns of fake news.
- *UPFD* [34] learns user preferences through their past engaged posts, and combines content with graph modeling.
- *MFAN* [17] integrates textual, visual, and social graph features in one unified framework for rumor detection.

3) Implementation Details: Following [32], all datasets are randomly split into training, validation and testing set with a ratio of 7:1:2. We employ AdamW [35] as the optimizer, and the batch size is set at 16. The initial learning rate is set to $5e^{-3}$. The hidden size of user embedding is set to 128. The The experiments are implemented based on PyTorch 1.13.0 on a Linux server equipped with GeForce RTX3090 GPUs.

B. Quantitative Evaluation

Each detection model is executed five times, with the average outcomes presented in Tables II and III. A comprehensive comparison of full propagation paths with propagation-enhanced methods is conducted, as shown in Table II. Our proposed *RPPG-Fake* achieves competitive results when compared with other propagation-enhanced baselines. Notably, *RPPG-Fake* surpasses all baselines on the PolitiFact and PHEME dataset across all evaluated metrics. This indicates that the generated propagation paths effectively provide additional contextual information, even in scenarios where data is sufficient. The improvement of detection capabilities is achieved through the incorporation of comprehensive propagation paths into propagation-enhanced methods. The propagation-enhanced approach Bi-GCN effectively leverages all propagation-path data during news dissemination, resulting in the optimal performance on the GossipCop dataset. Notably, the propagation paths within the GossipCop dataset provide abundant supplementary social context information for fake news detection. While an increase in generated propagation paths may not necessarily introduce additional social context for detection, our proposed method still achieves the second-best performance on precision metric on the GossipCop dataset. This finding suggests that the generation path is particularly effective in data-sparse scenarios and can enhance detection accuracy even when sufficient propagation path information is available.

To assess the detection efficacy of *RPPG-Fake* at early stages of news propagation, comparative analyses with early detection benchmarks are presented in Table III. For the simulation of early detection scenarios within the test dataset, early user engagements are retained randomly within the range of three to five as the accessible propagation paths. It is evident that the performance of propagation-enhanced methods significantly deteriorates, at times falling below that of some content-based methods such as CAFE. This suggests that a limited number of propagation paths struggle to capture the complex global structures and temporal information effectively during news spreading. Further comparisons are made with large language model (LLM) baselines such as LLAMA2 [36] and Detect-GPT [37], which utilize zero-shot prompt engineering. Interestingly, these LLM-based approaches underperform compared to both propagation-enhanced methods (in Table II) and our *RPPG-Fake*, indicating that textual features alone are inadequate for improving the accuracy of detection. The integration of generated propagation paths by *RPPG-Fake* markedly enhances detection capabilities, underscoring the critical importance of ample propagation paths. Notably, *RPPG-Fake* consistently surpasses all baselines across three datasets, registering accuracy improvements of +4.0% on PolitiFact, +4.1% on GossipCop, and +2.7% on Pheme. Our proposal *RPPG-Fake*, which generates propagation paths from news content, presents minimal social hazards while demonstrating superior performance at early stage detection compared to propagation-enhanced methods.

C. Impact of Propagation Paths

Fig. 3 illustrates the impact of propagation paths with four distinct methods across three datasets, simulating the progression from the early detection stage to a full propagation paths stage. The observed trend indicates an improvement in detection performance with the increase of propagation paths across all propagation-enhanced methods. This observation demonstrates the integrated value of extensive propagation paths in furnishing supplementary social context information, thereby enhancing the efficacy of fake news detection. Notably, in contrast to other propagation-enhanced models such as MFAN and Bi-GCN, our proposed RPPG-Fake exhibits minimal variation with the increase of propagation paths. This observation suggests that the generated propagation paths closely resemble real paths and consequently contribute valuable social context information to the detection process. With the full propagation paths and generation paths, our proposal achieves the highest detection performance. It indicates that generated propagation paths could provide external social context even in the situation of sufficient paths. This discovery underscores the efficacy of our approach in

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TABLE II
PERFORMANCE WITH FULL PROPAGATION PATHS

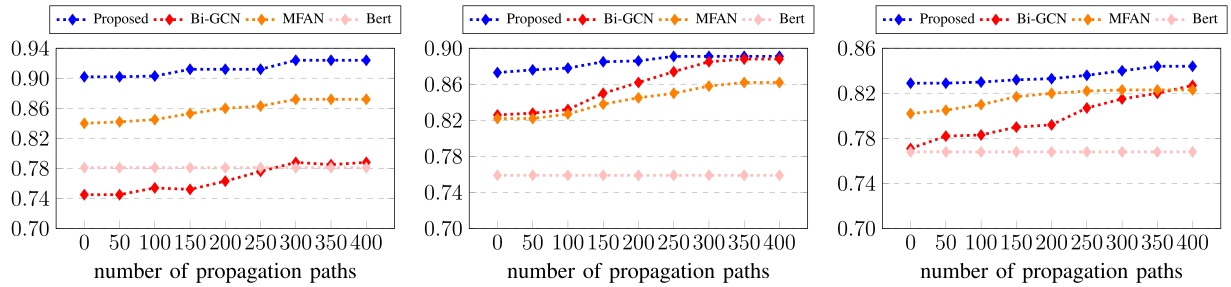
Category	Method	PolitiFact				GossipCop				Pheme			
		Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1
Propagation Enhanced	CSI	0.753	0.834	0.742	0.758	0.782	0.823	0.783	0.808	0.771	0.825	0.831	0.828
	Bi-GCN	0.788	0.834	0.742	0.758	0.890	**0.928**	0.848	0.890	0.827	0.840	0.834	0.835
	UPFD	0.904	0.923	0.857	0.889	0.879	0.906	0.931	0.918	–	–	–	–
	MFAN	0.872	0.841	0.881	0.860	0.862	0.903	0.910	0.906	0.823	0.878	0.847	0.862
Proposed	**RPPG-Fake**	**0.924**	**0.925**	**0.902**	**0.914**	**0.891**	0.911	**0.940**	**0.925**	**0.844**	**0.891**	**0.868**	**0.879**

Best results are in bold and second best results are underlined.

TABLE III
PERFORMANCE WITH FEWER PROPAGATION PATHS AT EARLY STAGE

Category	Method	PolitiFact				GossipCop				Pheme			
		Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1
Content Based	BERT	0.781	0.766	0.878	0.818	0.759	0.787	0.764	0.775	0.768	0.704	0.778	0.739
	MVAE	0.812	0.803	0.835	0.819	0.782	0.802	0.751	0.776	0.776	0.734	0.722	0.728
	EANN	0.804	0.808	0.794	0.781	0.796	0.812	0.765	0.788	0.771	0.714	0.701	0.704
	CAFE	0.848	0.857	0.850	0.853	0.832	0.804	0.903	0.851	–	–	–	–
	LLAMA2	0.848	0.865	0.780	0.820	0.825	0.811	0.872	0.840	–	–	–	–
	Detect-GPT	0.858	0.888	0.780	0.831	0.831	0.833	0.884	0.858	–	–	–	–
Propagation Enhanced	CSI	0.734	0.774	0.571	0.658	0.753	0.748	0.764	0.756	0.754	0.799	0.839	0.819
	Bi-GCN	0.745	0.696	0.762	0.727	0.826	0.871	0.881	0.876	0.771	0.825	0.831	0.828
	UPFD	0.862	0.892	0.786	0.835	0.822	0.860	0.891	0.875	–	–	–	–
	MFAN	0.840	0.846	0.787	0.815	0.822	0.861	0.888	0.875	0.802	0.853	0.846	0.850
Proposed	**RPPG-Fake**	**0.902**	**0.921**	**0.853**	**0.886**	**0.873**	**0.904**	**0.921**	**0.913**	**0.829**	**0.881**	**0.854**	**0.867**
	Imp.(%)	*+4.0*	*+2.9*	*+0.3*	*+3.3*	*+4.1*	*+3.3*	*+1.8*	*+3.7*	*+2.7*	*+2.8*	*+0.9*	*+1.7*

Best results are in bold and second best results are underlined.



(a) PoltiFact (b) GossipCop (c) Pheme

Fig. 3. The accuracy performance with different generation users. Comparison of the proposed RPPG-Fake model with baselines across three datasets.

generating high-quality propagation paths at the early detection stage, thereby integrating substantial social context into fake news detection and mitigating the potential societal hazards stemming from the dissemination of fake news.

D. Ablation Study

The ablation studies are conducted to investigate the importance of different modules in Table IV.

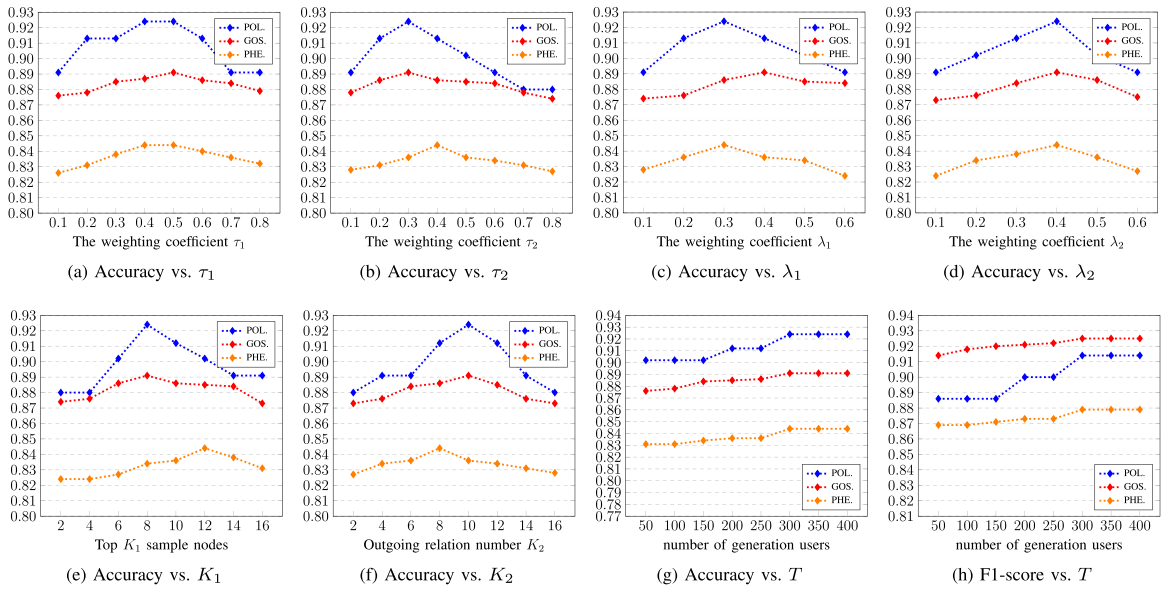
Heterogeneous Contrastive Learning: We aim to study the influence of various contrastive learning methods. "w/o Node-view CL" and "w/o Meta-path CL" stand for without node-view contrastive learning, and meta-path contrastive learning. "w/o all CL" stand for without all contrastive learning and initialize with random user embedding. (1) The performance declines with "w/o Node-view CL" or "w/o Meta-path CL", revealing that propagation information is as important as heterogeneous relations for user embedding in social networks. (2) Without all contrastive learning, performance declines significantly, proving both contrastive learning are crucial to generate the reasonable user embedding.

Propagation Path Generation: Here we focus on investigating the effectiveness of reinforced propagation path generation. "Random Sample" denotes to sample 10 user nodes as the neighbors at current time step. "Node Similarity" is designed to select top-10 context neighbors based on the semantic similarities. (1) After replacing the reinforced propagation path generation with random samples, model performance presents a significant decline, verifying that random samples may introduce noises and reasonable user interaction is crucial to improve

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TABLE IV
ABLATION STUDIES OF *RPPG-FAKE* ON THREE DATASETS

Abaltion Categoty	Method	PolitiFact				GoosipCop				Pheme			
		Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1
Proposed	RPPG-Fake	**0.924**	**0.925**	**0.902**	**0.914**	**0.891**	**0.911**	**0.940**	**0.925**	**0.844**	**0.891**	**0.868**	**0.879**
Heterogeneous Contrastive Learning	w/o all CL	0.891	0.919	0.829	0.872	0.874	0.910	0.921	0.915	0.827	0.879	0.852	0.867
	w/o Node-view CL	0.902	0.921	0.854	0.886	0.876	0.905	0.924	0.914	0.828	0.881	0.852	0.867
	w/o Meta-path CL	0.913	0.902	0.902	0.902	0.885	0.907	0.936	0.921	0.831	0.883	0.856	0.869
Propagation Path Generation	Random Sample	0.891	0.897	0.854	0.875	0.872	0.907	0.917	0.912	0.824	0.879	0.849	0.864
	Node Similarity	0.902	0.921	0.854	0.886	0.873	0.905	0.921	0.913	0.826	0.881	0.850	0.865
Temporal-aware Detection	w/o structure feature	0.913	0.923	0.878	0.900	0.886	0.908	0.937	0.922	0.834	0.885	0.858	0.871
	w/o temporal feature	0.902	0.900	0.878	0.889	0.884	0.906	0.935	0.920	0.836	0.887	0.859	0.873



(a) Accuracy vs. τ_1 (b) Accuracy vs. τ_2 (c) Accuracy vs. λ_1 (d) Accuracy vs. λ_2

(e) Accuracy vs. K_1 (f) Accuracy vs. K_2 (g) Accuracy vs. T (h) F1-score vs. T

Fig. 4. Parameter Sensitivity Analysis. Impact of different hyperparameters on accuracy and F1-score across three datasets.

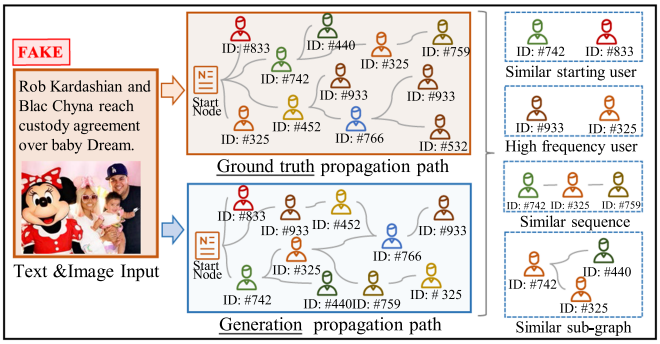
performance. (2) Compared with the manually defined selection strategy "Node Similarity", our proposal achieves better performance, revealing that the learning of propagation path is capable of capturing the news background-aware propagation patterns.

Temporal-aware Detection: Here we focus on the influence of temporal-aware fake news detection. "w/o structure feature" and "w/o temporal feature" are implemented without global propagation graph and dynamic propagation graph. (1) The performance declines more with 'w/o temporal feature", revealing that temporal information in propagation paths is more important than global structure feature. (2) Both modules are crucial to achieving desirable detection performance as both "'w/o structure feature" and "'w/o temporal feature"' only achieve degraded results. It demonstrates both global structure feature and temporal depth feature are beneficial for detection performances.

E. Parameter Sensitivity Analysis

Here we conduct hyper-parameter sensitivity analysis on six parameters: the temperature parameter τ_1 in node-view contrastive learning, temperature parameter τ_2 in meta-view contrastive learning, the weighting coefficient λ_1 in contrastive loss, the weighting coefficient λ_2 in reward loss, the topK_1 sample nodes in contrastive learning and the max number outgoing relations K_2 for one action in reinforcement learning. In Fig. 4(a) and (b), with the increase of τ_1, model performance first increases and then decreases. With a greater temperature parameter the similarity distribution is smoother, which is benefit for model capturing the minor difference. While the common feature may be ignored if the temperature parameter is too high. In Fig. 4(c), with the increase of λ_1, model performance first increases and then significantly drops. A greater λ_1 leads to more node view features with limited heterogeneous relation features, while node view global structure is decreased if λ_1 is too small. In Fig. 4(d), with the increase of λ_2, model performance improves and then decreases. A greater λ_2 leads to more temporal reward in every action step, while global propagation reward is introduced more if λ_2 is too small. Fig. 4(e) shows that with the increase of sample nodes K_1, model performance first improves and then drops. This demonstrates that the reasonable number of positive and negative node set are crucial for effective contrastive learning. Fig. 4(f) shows that with the increase of the max number of outgoing relations K_2, model performance first improves and then drops. This demonstrates that too much user interaction in one step may hurt the quality of generated propagation paths.

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(a) The propagation paths generation case on GossipCop dataset.

	Model	Input	Output	Feature
Content-based	MVAE		REAL	✗ Low Accuracy
Propagation Enhanced	Bi-GCN		FAKE	High Hazard
Propagation Simulation	RPPG-Fake		FAKE	✓ High Accuracy ✓ Low Hazard

(b) The comparison of different fake news detection models.

Fig. 5. Case studies on GossipCop dataset.

We further evaluate detection performance on the various number of generation users in Fig. 4. With the increase of generation users, the model performance first increases and then levels off. This demonstrates that the performance does not increase after a certain amount of propagated information and the social context integrated from propagation paths is sufficient for detection.

F. Case Study

Fig. 5(a) illustrates a generated propagation path by *RPPG-Fake*. The resemblance of these paths to the actual ground-truth is apparent, particularly in aspects such as the presence of high-frequency users, the identity of the initial spreader, the sequence of propagation, and the structure of the sub-graph. This resemblance indicates that our model effectively captures the key temporal and structural characteristics of news dissemination within user communities. We also conduct a comparative analysis of the detection process employing various methods, as illustrated in Fig. 5(b). Contrasted with the content-based approach, the RPPG-Fake generates the propagation path and enhances detection accuracy. In comparison with propagation-enhanced methods, our proposed utilizes news content as input without relying on real propagation paths. This observation signifies that our proposal enhances detection performance while concurrently mitigating social hazards.

V. Related Work

A. Related Work

Content-based: In recent years, there has been widespread attention to the automated detection of fake news in social media. Some early research endeavors seek to detect fake news by extracting features from the textual content of the news articles [2], [38], [39], [40], [41], [42]. For instance, Wawer et al. [43] utilize textual and linguistic features from websites, based on bag-of-words vector space and psycholinguistic dimensions, to predict the credibility of websites. Vaibhav et al. [44] propose a model based on graph neural networks to capture the interaction among sentences in the content of fake news. Meanwhile, some researchers propose novel approaches utilizing cross-modal features in news to enhance the accuracy of fake news detection [4], [32], [45], [46], [47]. Khattar et al. [1] propose the MVAE model which consists of three components: an encoder, a decoder, and a fake news classifier. The encoder is used to encode a shared representation of features, the decoder reconstructs the data from multi-modal representations, and the fake news classifier categorizes news into true or false categories. Zhou et al. [48] propose the SAFE method by computing the correlation between textual information and visual information, defining it as a modified cosine similarity to detect fake news. Wu et al. [49] uses multiple co-attention layers to learn the relationship between text and images.

Recently, researchers have started exploring the utilization of Large Language Models (LLMs) for fake news detection. Some studies focus on directly prompting various LLMs such as GPT-3 [37], ChatGPT-3.5 [50], [51] and GPT-4 [52] for misinformation detection. For instance, Pan et al. [53] introduce a program-guided fact-checking framework leveraging the contextual learning ability of LLMs to generate reasoning programs guiding veracity verification. Wu et al. [54] apply GPT-3.5 as a feature extractor to identify out-of-context images. However, existing LLM-based fake news detection methods primarily rely on textual semantics, often insufficient for effectively considering user behaviors during news dissemination. Textual features alone may not adequately verify the veracity of news items in certain situations.

Propagation-enhanced: Different from methods that rely on the content of news for fake news detection, propagation-enhanced fake news detection approaches aim to improve accuracy by leveraging differences in the propagation processes between real and fake news [10], [55], [56], [57], [58]. Jin et al. [59] apply epidemiological models to characterize information cascades triggered by both real and fake news on Twitter. Ma et al. [60] use a recursive neural network to represent features of news by integrating both the propagation structure and news features. Zhang et al. [61] propose a deep diffusive Network model that can simultaneously learn latent representations and infer the accuracy of news articles, creators, and topics. Liu et al. [9] conduct an authenticity assessment of news based on user profile information within the news propagation network. Dou et al. [34] integrate user past engaged posts into news content with graph modeling. Zheng et al. [17] combine textual, visual, and social graph features with cross-attention for rumor detection. Wei et al. [62] design a structure adversarial net framework to learn transferable features from propagation to solve the problem of cold-start fake news detection. Lakzaei et al. [63] adopt a graph-based model and introduce a semi-supervised and one-class approach to overcome the limited availability of labeled news datasets.

Due to the potential impact of misinformation on a large audience and its negative consequences during dissemination,

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the early detection of fake news has become a crucial research focus within the field of fake news detection [13], [64], [65]. Yang et al. [66] endeavor to utilize convolutional neural networks for extracting linguistic and user features from news content and employ these features for the early detection of fake news. Nguyen et al. [67] employ deep neural networks to capture features at a posterior level, achieving superior performance in fake news early detection. Liu et al. [68] propose a novel deep neural network that integrates crowd response features and user reactions. Yuan et al. [14] propose to combine news content, publishing, and reposting relations for early detection. Huang et al. [13] design a social bot-aware graph neural network for early detection.

VI. Conclusion

This paper introduces the innovative *RPPG-Fake* model. The model simulates the propagation process using the contents of news and integrates the generated propagation paths into a temporal-aware detection module. The effectiveness of the proposed model is thoroughly assessed across three popular datasets. The experimental results demonstrate the enhanced performance of our proposal. While the model shows promise, future work could explore more sophisticated propagation path generation techniques that account for additional social dynamics, incorporate adaptive learning for better generalization across domains, and focus on optimizing real-time detection and scalability for large-scale social networks. Addressing these areas will be essential to maximizing the model's impact and applicability across diverse contexts.

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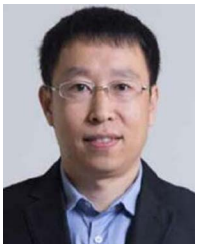


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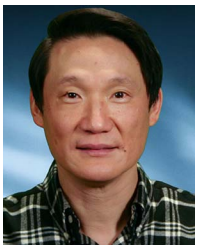


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